Retrained-model high-SNR safeguards: identical magnitude-domain data and detector

2 paths/user

4 paths/user

3–7 paths/user

RSR 5 dB Uncoded BER

RSR 12 dB Uncoded BER

RSR 20 dB Uncoded BER

$10^{-1}$ $10^{-2}$ $10^{-3}$ $10^{-4}$ $10^{-5}$ $10^{-6}$

0 20 30 40 50 60

SNR (dB)

Perfect CSI

Retrained filtered

Calibrated EM100 bypass

Same-step bypass diagnostic

Direct EM100

Fresh compact sparse

Zero observed errors (marker only)

Each channel uses all64 jointQPSK symbols once. Zero-error markers at$10^{-6}$ are display markers, not estimates or bounds.